



PROJECT ENGINEERING TRAINEE • CAMPUS HIRING

What e-con Systems is Expecting from You

A preparation guide for the Hardware and Software Trainee interview process



Hardware Trainee

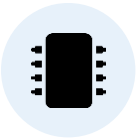


Software Trainee



Two Trainee Tracks, One Hiring Process

Every candidate is evaluated for one of two Project Engineering Trainee roles. Both tracks value strong fundamentals, hands-on application, and clarity about your own project work.



Project Engineering Trainee — Hardware

- Analog Electronics
- Digital Electronics
- Circuit Theory & Network Analysis
- Basic Knowledge of Electricals

Circuits + applications, not just theory



Project Engineering Trainee — Software

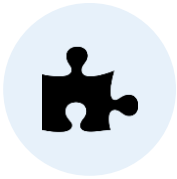
- Logical Puzzle Assessment (first round)
- Reasoning, Maths, Aptitude & Analytical Puzzles
- Embedded C — added advantage
- Depth in your own project work

Reasoning first, coding language second



The Logical Puzzle Round

All Software candidates begin with the same Logical Puzzle assessment — this is the common gate before anything else.



Every candidate attends this assessment first — performance here determines who moves forward in the process.

Topics covered in the assessment

- Logical Reasoning (Verbal & Non-Verbal)
- Geometrical Problems
- Data Interpretation
- Algebraic Problems
- Mathematical Problem Modelling
- Time, Speed & Distance / Time & Work
- Practical & Analytical Puzzles
- Probability & Combinatorics
- Statistics
- Physics & Physical Properties



What the Questions Actually Look Like

1

Probability — Geometry

A point is chosen uniformly at random inside a unit square. What is the probability that the point lies within 0.5 units from the centre of the square?

2

Time & Work — Modelling

A team had to lay tiles on two fields. One field was twice the size of the other. The team worked for half a day on the larger field, then split into two equal groups. The first group stayed and finished the larger field by evening. The second group tiled the smaller field but left a portion unfinished by evening — a portion that one person completed the next day in a single day's work. How many people were in the team?



Coding Skills & Your Project Story



Programming Strength

- Embedded C is an added advantage
- If you know other languages, be genuinely strong in them — depth matters more than breadth
- Be ready to write and reason through code live, not just describe it



Know Your Projects Cold

- Your exact contribution to the project
- What you learned building it
- What motivated you to take it on
- The final output / result achieved



HARDWARE TRACK

Come Prepared — In Depth

The Hardware round spans four subject areas. Interviewers expect more than definitions.



Don't just focus on theory — work out circuits and real applications



Practice numericals: apply each theorem/law as a solvable sum, not just a definition



Prepare all four topic areas in depth — questions can come from any of them



Know your own projects thoroughly: contribution, learnings, and outcome



Analog Electronics



Focus on biasing conditions, operating regions, and how each circuit behaves under real input — not just circuit names.

Be ready to work through op-amp gain, clipper/clamper output waveforms, and rectifier ripple calculations.

- BJT Biasing and Region of BJT
- MOSFET Biasing
- Operational Amplifier
- Zener Diode as Regulator
- Clipper Circuit
- Clamper Circuit
- Rectifier & Filter
- Low Frequency BJT Amplifier



Digital Electronics



Practice truth tables, K-map simplification, and number system conversions by hand.

Be comfortable tracing 8085 instruction execution and comparing logic family characteristics (TTL vs CMOS).

- Logic Gates
- Combinational Circuits
- Number Systems, Binary Codes & Complement Form
- Sequential Circuits
- DAC & ADC
- Microprocessor 8085
- Logic Family
- Boolean Algebra



Circuit Theory

The largest topic area — expect direct problem-solving. Every theorem listed should be practiced as a numerical, not just recalled as a concept.

- Basic Concepts of Networks (Voltage, Current, Power, etc.)
- Ohm's Law — Problem Solving
- Kirchhoff's Laws — KVL & KCL
- Voltage Division & Current Division Rule
- Finding Equivalent Resistance
- Nodal Analysis and Problem Solving
- Star to Delta Conversion ($Y-\Delta$)

- Thevenin's Theorem
- Norton's Theorem
- Superposition Theorem
- Maximum Power Transfer Theorem
- DC Transient Analysis
- Resonance Circuits



Basic Knowledge of Electricals



Round out your preparation with the fundamentals of electrical machines and power systems.

Expect questions that start from basic transformer theory and move quickly into numericals.

- Transformer Basics
- Working Principle & Construction
- EMF Equation
- Turns Ratio & Voltage/Current Relations
- Losses and Efficiency
- Types of Transformers



BEFORE YOU WALK IN

Three Things to Get Right



Depth over breadth

Whichever role you're targeting, prepare every listed topic thoroughly — questions can come from anywhere in the list.



Apply, don't just recall

Work through circuits, sums, and worked problems for every theorem or concept — not just definitions.



Own your projects

Be ready to explain your contribution, what you learned, your motivation, and the final output in detail.

Good luck — we're looking forward to meeting you.